



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech. Course: Electrical Machine Design Course Code: PCC-EE403
Date: 20.08.2025 Time: 15:00 to 15:45 Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Attempt all questions.	Marks	CO	BT
Q.1	List out major differences between distribution and power transformer.	5	CO1	BT1
Q.2	Derive an expression for kVA output of single-phase transformer.	5	CO1	BT2
Q.3	A 3-Phase core type transformer of 400 kVA, 6600/440 V, 50 Hz, has 15 V emf per turn, 1.5 Wb/m^2 maximum flux density, 2 A/mm^2 current density, 0.2 window space factor, 0.9 stacking factor, use a 3 stepped core. For a three stepped core width of largest stamping = $0.9 d$, net iron area = $0.6 d^2$. where, d is the diameter of circumscribing circle. If transformer overall height is equal to overall width, then calculate approximate overall dimensions.	6	CO1	BT3
Q.4	a Explain the path of heat flow in a transformer with a neat diagram.	2	CO2	BT2
	b Find the specific heat dissipation due to convection of transformer oil if there is 15°C temperature difference of the surface relative to oil and 0.75 m height of dissipating surface. Also compare it with the specific heat dissipation due to radiation of air.	2	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the constructional details and design aspects of various types of transformers and calculate parameters such as EMF per turn, core dimensions, and insulation. (Understand, Apply)
CO 2	Evaluate transformer performance using equivalent circuit modeling, leakage reactance, and temperature rise analysis, and design cooling systems like tanks and radiators. (Evaluate, Design)
CO 3	Describe the construction and design process of three-phase induction motors including stator and rotor components, and calculate parameters like turns per phase, air gap length, and rotor currents.
CO 4	Analyze the performance of three-phase induction motors using circle diagram and derive key parameters such as maximum torque, power factor, and dispersion coefficient. (Analyze, Evaluate)
CO 5	Explain the construction and design methodology of synchronous machines and estimate main dimensions using electric/magnetic loadings and short-circuit ratio. (Understand, Apply)
CO 6	Apply computer-aided design techniques and optimization approaches to develop improved designs of transformers and induction motors. (Apply, Design)



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: B.Tech.

Course: High Voltage Engineering

Course Code: PEC-EE405

Date: 18.08.2025

Time: 11:00 AM to 11:45 AM

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	ATTEPT ANY FOUR	Marks	CO	BT
Q.1	Derive current growth equation in the presence of secondary ionization processes.	5	CO1	BT2
Q.2	Explain streamer theory of breakdown in air at atmospheric pressure.	5	CO1	BT2
Q.3	Draw Panchen's curve and Derive expression for Panchen's Law.	5	CO1	BT2
Q.4	In an experiment in a certain gas it was found that the steady state current is 5.5×10^{-8} A at 8 kV at a distance of 0.4 cm between the plane electrodes. Keeping the field constant and reducing the distance to 0.1 cm results in a current of 5.5×10^{-9} A. Calculate Townsend's primary ionization coefficient α .	5	CO1	BT3
Q.5	Describe the working of Van de Graaff generator with neat sketch.	5	CO2	BT2
Q.6	Explain with neat sketches and its operation of voltage doubler circuits.	5	CO2	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Observe the breakdown mechanism in gaseous, liquid and solid insulation.
CO 2	Illustrate the method of high voltage generation, impulse voltage and current.
CO 3	Describe the methods of Measurement High voltage, Impulse voltage and current.
CO 4	Design various circuits for the measurement of high frequency voltages and currents.
CO 5	Apply the different tests done on insulators, circuit breakers, cables, transformers LA etc.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering In-Semester Examination I Academic Year: 2025-26 (Semester – I)		
Class: Final Year	Subject: Switchgear and Protection		Subject Code: PCC EE 402
Date: 21.08.2025	Time: 11:00 to 11:45		Max. Marks: 20

Note:

- i. All Questions are compulsory.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/ Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. Question paper contains only 01 pages.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain the construction and working of an HRC fuse with the help of a neat diagram.	4	CO-1	BT-2
Q.2	Explain the following concepts related to circuit breakers: (a) Capacitive current breaking (b) Inductive current breaking (c) Resistance switching (d) Auto reclosing	8	CO-1	BT-2
Q.3	Draw and explain the construction and working of a Minimum Oil Circuit Breaker	4	CO-1	BT-2
Q.4	Define following terms related to Arc (a) Arc voltage (b) Restriking voltage (c) Recovery voltage (d) RRRV.	4	CO-1	BT-1

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Knowledgeable in field of power system protection, circuit breakers, protective relaying and instrument transformer
CO 2	Comprehensive study of various relays used in power system protection
CO 3	Discuss types of circuit breakers with their applications
CO 4	Identify rotor, stator faults, interterm faults and their protection